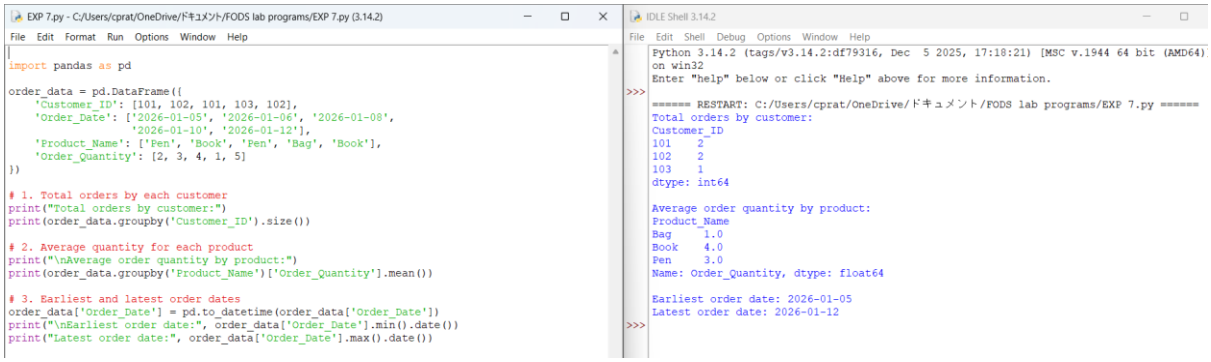
Fundamentals of Data Science Lab Programs Set (7 – 15)

7. Analyze customer orders using Pandas

Program:

```
import pandas as pd
order_data = pd.DataFrame({
    'Customer_ID': [101, 102, 101, 103, 102],
    'Order_Date': ['2026-01-05', '2026-01-06', '2026-01-08',
                   '2026-01-10', '2026-01-12'],
    'Product_Name': ['Pen', 'Book', 'Pen', 'Bag', 'Book'],
    'Order_Quantity': [2, 3, 4, 1, 5]
})
print("Total orders by customer:")
print(order_data.groupby('Customer_ID').size())
print("\nAverage order quantity by product:")
print(order_data.groupby('Product_Name')['Order_Quantity'].mean())
order_data['Order_Date'] = pd.to_datetime(order_data['Order_Date'])
print("\nEarliest order date:", order_data['Order_Date'].min().date())
print("Latest order date:", order_data['Order_Date'].max().date())
```

OUTPUT



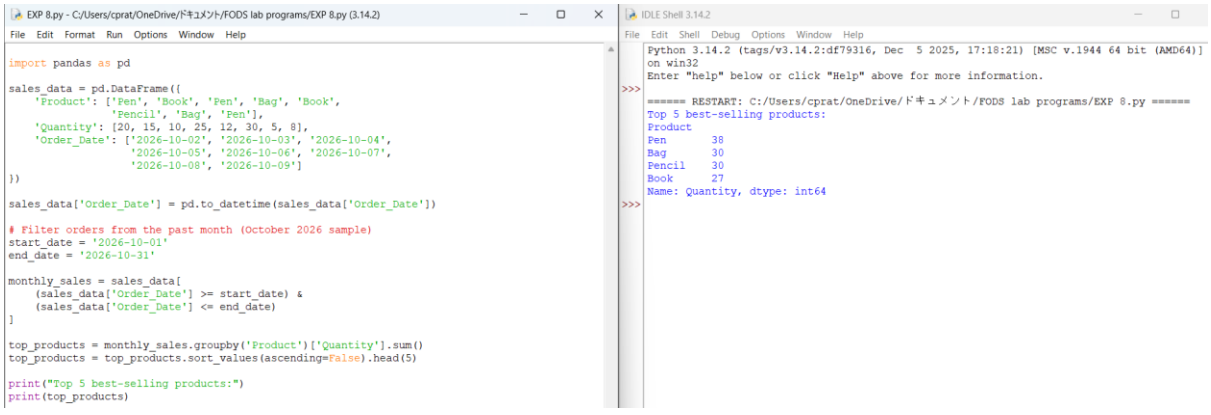
8. Find the top 5 best-selling products in the past month

Program:

```
import pandas as pd
```

```
sales_data = pd.DataFrame({
    'Product': ['Pen', 'Book', 'Pen', 'Bag', 'Book',
                'Pencil', 'Bag', 'Pen'],
    'Quantity': [20, 15, 10, 25, 12, 30, 5, 8],
    'Order_Date': ['2026-10-02', '2026-10-03', '2026-10-04',
                   '2026-10-05', '2026-10-06', '2026-10-07',
                   '2026-10-08', '2026-10-09']
})
sales_data['Order_Date'] = pd.to_datetime(sales_data['Order_Date'])
start_date = '2026-10-01'
end_date = '2026-10-31'
monthly_sales = sales_data[
    (sales_data['Order_Date'] >= start_date) &
    (sales_data['Order_Date'] <= end_date)
]
top_products = monthly_sales.groupby('Product')['Quantity'].sum()
top_products = top_products.sort_values(ascending=False).head(5)
print("Top 5 best-selling products:")
print(top_products)
```

OUTPUT



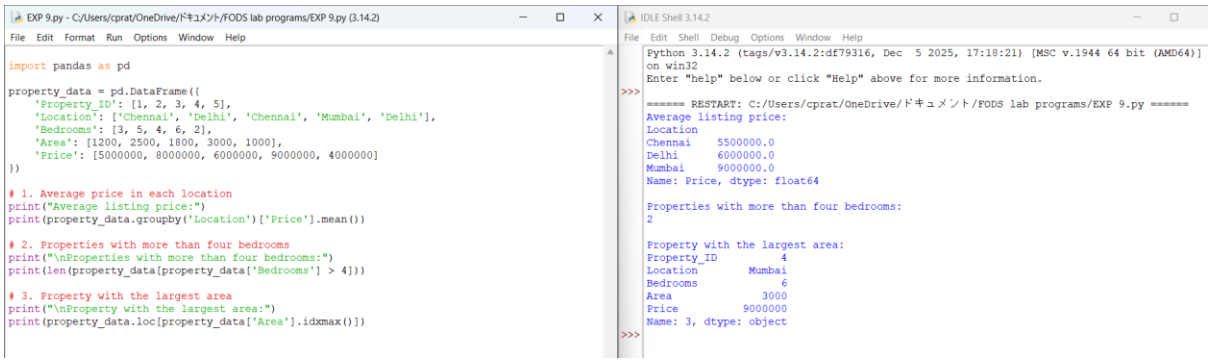
9. Analyze real estate properties using Pandas

Program:

```
import pandas as pd
```

```
property_data = pd.DataFrame({
    'Property_ID': [1, 2, 3, 4, 5],
    'Location': ['Chennai', 'Delhi', 'Chennai', 'Mumbai', 'Delhi'],
    'Bedrooms': [3, 5, 4, 6, 2],
    'Area': [1200, 2500, 1800, 3000, 1000],
    'Price': [5000000, 8000000, 6000000, 9000000, 4000000]
})
print("Average listing price:")
print(property_data.groupby('Location')['Price'].mean())
print("\nProperties with more than four bedrooms:")
print(len(property_data[property_data['Bedrooms'] > 4]))
print("\nProperty with the largest area:")
print(property_data.loc[property_data['Area'].idxmax()])
```

OUTPUT



10. Create line and bar plots for monthly sales

Program:

```
import matplotlib.pyplot as plt
months = ['Jan', 'Feb', 'Mar', 'Apr', 'May']
sales = [100, 150, 120, 200, 250]
plt.plot(months, sales, marker='o')
plt.title("Monthly Sales - Line Plot")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.show()
```

plt.bar(months, sales)

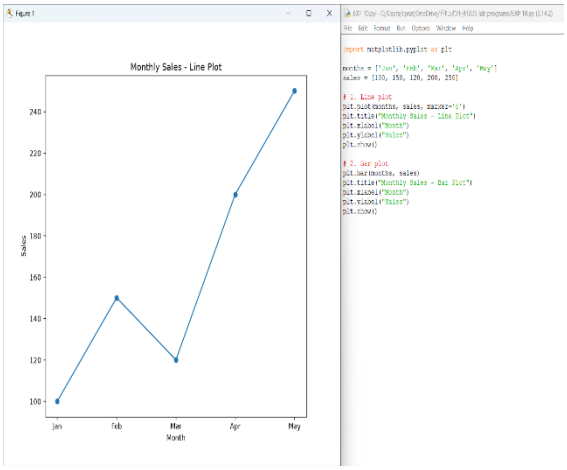
plt.title("Monthly Sales - Bar Plot")

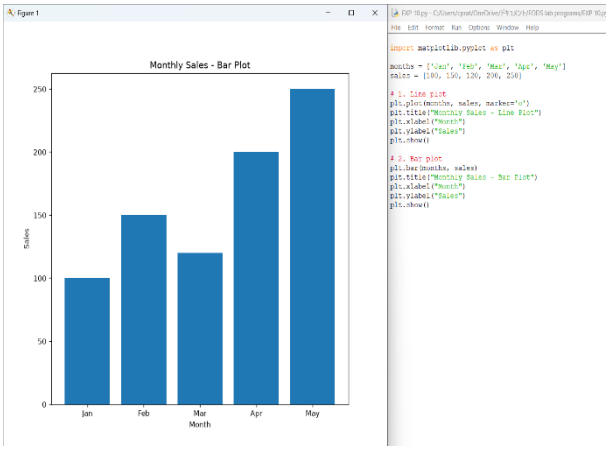
plt.xlabel("Month")

plt.ylabel("Sales")

plt.show()

OUTPUT





11. Create line, scatter, and bar plots for monthly sales

Program:

import matplotlib.pyplot as plt

months = ['Jan', 'Feb', 'Mar', 'Apr', 'May']

sales = [100, 150, 120, 200, 250]

plt.plot(months, sales, marker='o')

plt.title("Monthly Sales - Line Plot")

plt.xlabel("Month")

plt.ylabel("Sales")

plt.show()

plt.scatter(months, sales)

plt.title("Monthly Sales - Scatter Plot")

plt.xlabel("Month")

plt.ylabel("Sales")

plt.show()

plt.bar(months, sales)

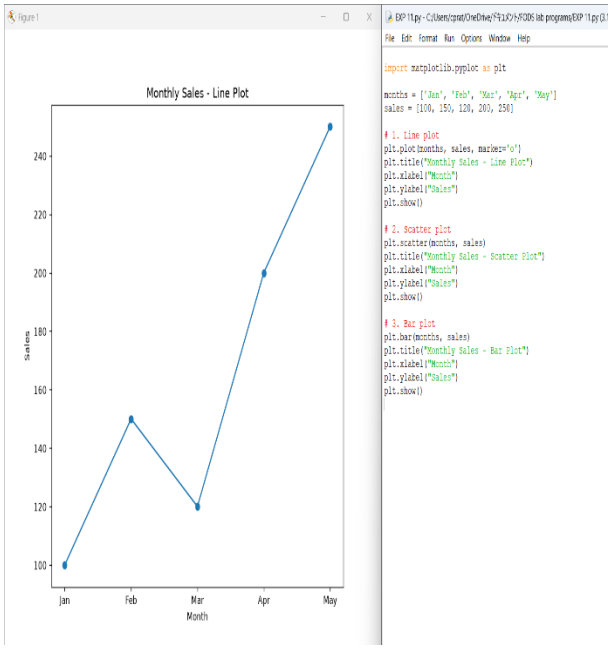
plt.title("Monthly Sales - Bar Plot")

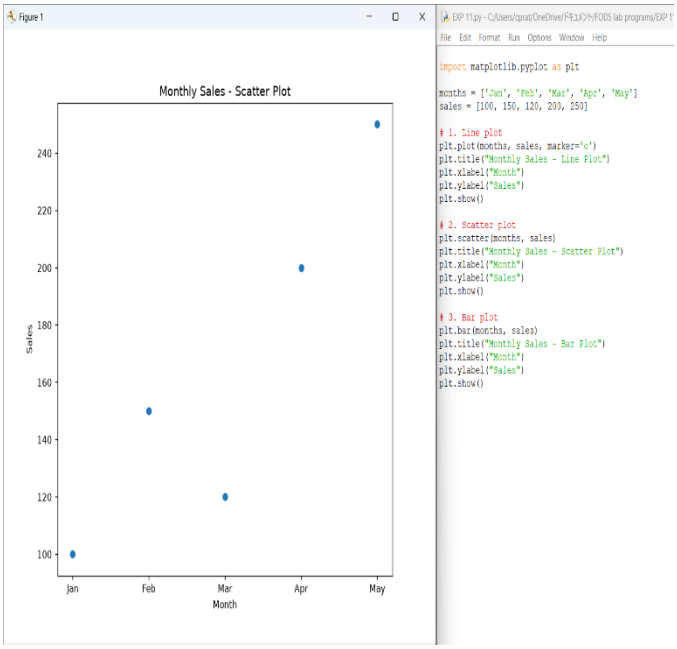
plt.xlabel("Month")

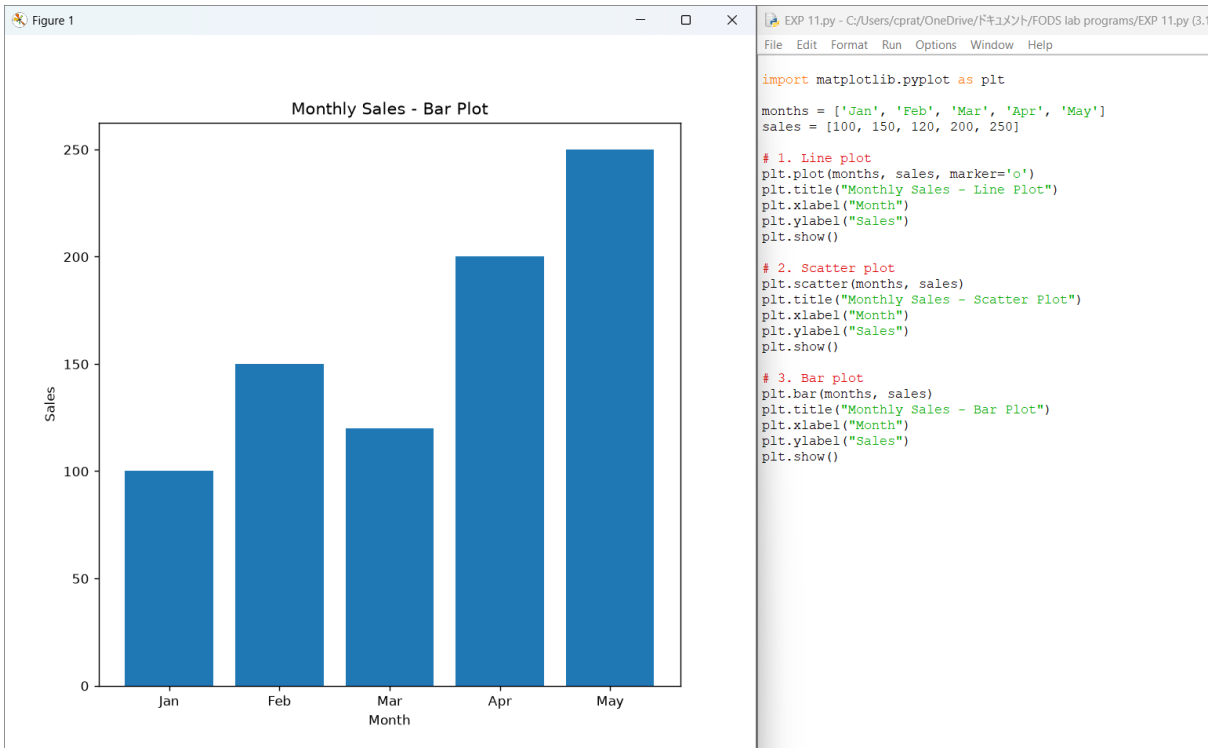
plt.ylabel("Sales")

plt.show()

OUTPUT





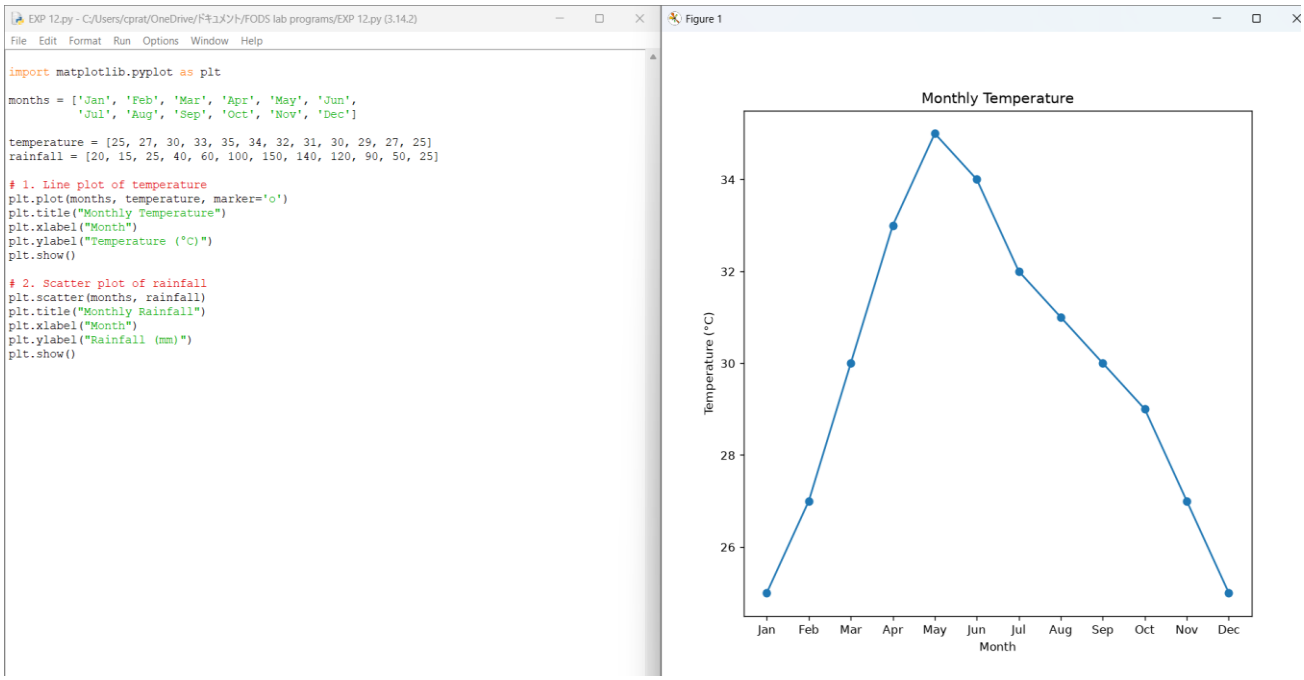


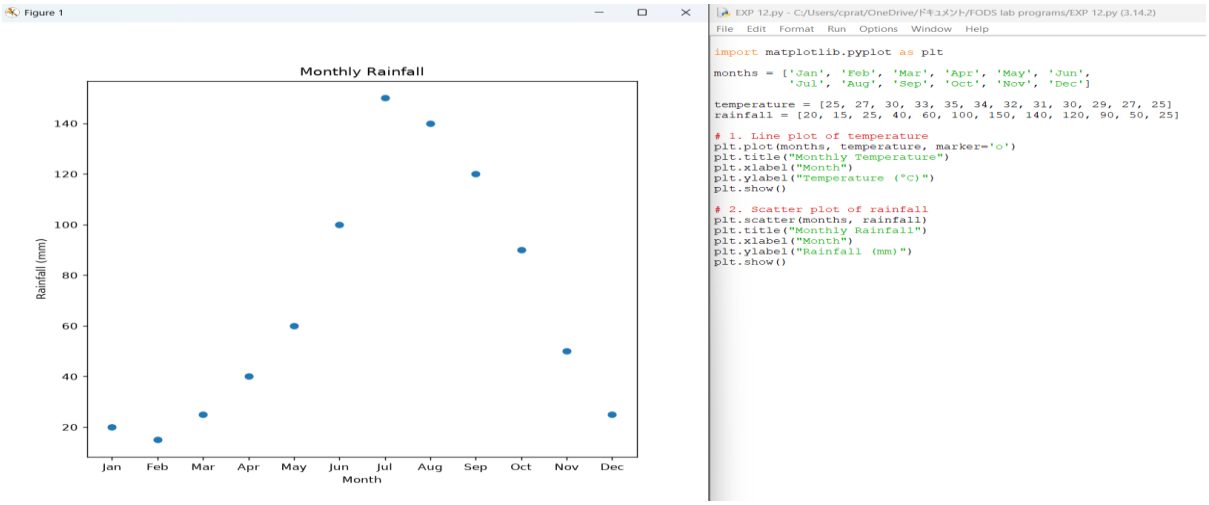
12. Create line plot for temperature and scatter plot for rainfall

Program:

```
import matplotlib.pyplot as plt
months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun',
        'Jul', 'Aug', 'Sep', 'Oct', 'Nov', 'Dec']
temperature = [25, 27, 30, 33, 35, 34, 32, 31, 30, 29, 27, 25]
rainfall = [20, 15, 25, 40, 60, 100, 150, 140, 120, 90, 50, 25]
plt.plot(months, temperature, marker='o')
plt.title("Monthly Temperature")
plt.xlabel("Month")
plt.ylabel("Temperature (°C)")
plt.show()
plt.scatter(months, rainfall)
plt.title("Monthly Rainfall")
plt.xlabel("Month")
plt.ylabel("Rainfall (mm)")
plt.show()
```

OUTPUT





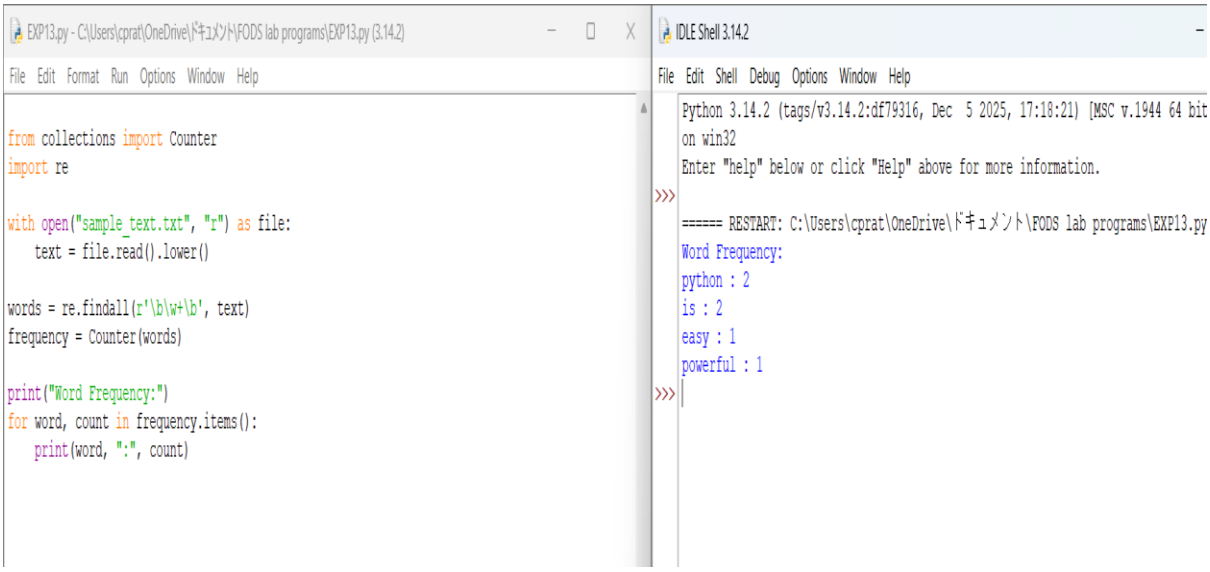
13. Calculate word frequency from a text file

Program:

Create a text file named sample_text.txt containing:

```
from collections import Counter
import re
with open("sample_text.txt", "r") as file:
    text = file.read().lower()
words = re.findall(r'\b\w+\b', text)
frequency = Counter(words)
print("Word Frequency:")
for word, count in frequency.items():
    print(word, ":", count)
```

OUTPUT

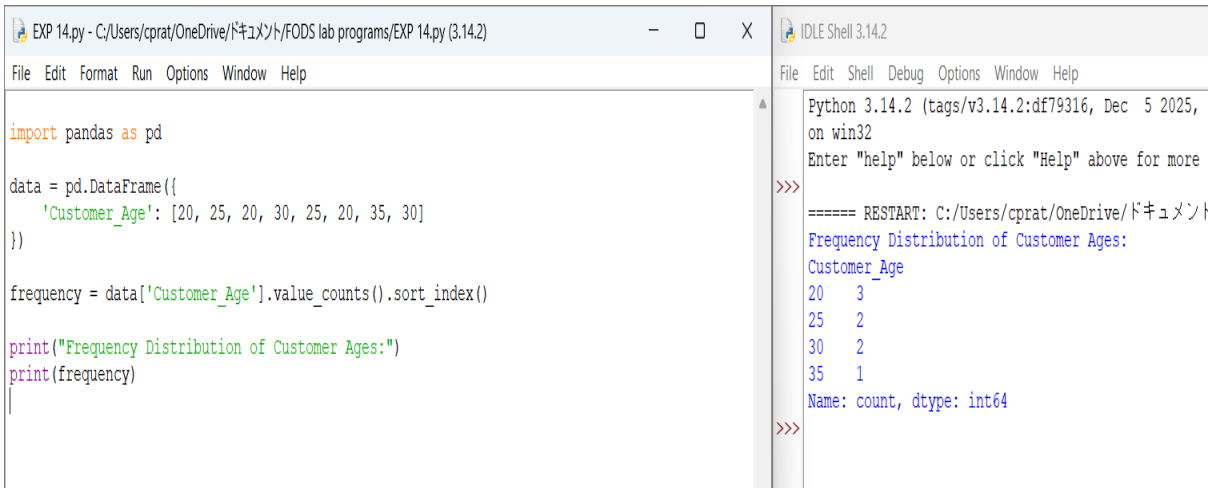


14. Find the frequency distribution of customer ages

Program:

```
import pandas as pd
data = pd.DataFrame({
    'Customer_Age': [20, 25, 20, 30, 25, 20, 35, 30]
})
frequency = data['Customer_Age'].value_counts().sort_index()
print("Frequency Distribution of Customer Ages:")
print(frequency)
```

OUTPUT



15. Calculate the frequency distribution of likes among posts

Program:

```
import pandas as pd
data = pd.DataFrame({ 'Post_ID': [1, 2, 3, 4, 5, 6, 7, 8],
    'Likes': [100, 200, 100, 300, 200, 100, 400, 300]})
frequency = data['Likes'].value_counts().sort_index()
print("Frequency Distribution of Likes:")
print(frequency)
```

OUTPUT

